HUAWEI PRACTICE COMPETITION 2025/2026

Cloud Track — Master Practice Exam & Complete Solutions Guide

Total Questions: *150* | ***Total Marks:*** *1,000 Marks* | ***Domains:*** *Cloud Infrastructure (90 Qs) & Artificial Intelligence (60 Qs)*
Includes: *Part 1 — Full Exam Paper • Part 2 — Master Answer Key Matrix & Official Solutions Guide*

PART 1: PRACTICE EXAM PAPERS

Section 1: Single-Answer Questions (Q1 – Q70) [560 Marks]

Question 1 [8 Marks]

Which of the following load balancing algorithms is not supported by shared and dedicated load balancers?

A. Weighted round robin
B. Source IP hash
C. Destination IP address
D. Weighted least connections

Question 2 [8 Marks]

Which of the following is not a requirement for creating a Virtual Private Cloud (VPC) peering connection?

A. The two VPCs must be in the same region.
B. The CIDR blocks of the two VPCs cannot overlap.
C. The two VPCs must belong to the same project.
D. Routes must be configured for both VPCs.

Question 3 [8 Marks]

An enterprise connects its on-premises network to a VPC through a Direct Connect connection. As the business grows, they need another VPC in the same region, which also needs to communicate with the on-premises network. To reduce costs, the enterprise wants to reuse the existing Direct Connect connection. Which of the following solutions is not cost-effective?

A. The enterprise can buy a VPN connection to connect the VPCs to the on-premises network.
B. The enterprise needs to add the CIDR block of the new VPC to the local subnet of the Direct Connect connection.
C. The enterprise can create a VPC peering connection between the two VPCs, and add required routes.
D. The enterprise needs to add routes on the on-premises gateway with the destination pointing to the VPCs.

Question 4 [8 Marks]

A company plans to deploy applications in the CN South-Guangzhou and CN East-Shanghai regions and uses Cloud Connect to enable network communications between the two regions. Which of the following statements about Cloud Connect is false?

A. The subnets of the VPCs loaded to a cloud connection cannot conflict with each other.
B. Cloud Connect can connect VPCs of different accounts.
C. Cloud Connect can connect VPCs in different regions.
D. Cloud Connect can only connect VPCs of the same account.

Question 5 [8 Marks]

Which of the following statements about Elastic Load Balance (ELB) listeners is false?

A. A listener checks user requests and routes traffic to backend server groups based on configured rules.
B. A load balancer can have multiple listeners configured.
C. Listeners support Layer 4 (TCP/UDP) and Layer 7 (HTTP/HTTPS) protocols.
D. A listener can only be bound to one backend server group.

Question 6 [8 Marks]

Which of the following is an advantage of Virtual Machines (VMs) over containers?

A. Fine granularity and high resource utilization
B. Startup in seconds
C. Lightweight with fewer resources
D. Better isolation

Question 7 [8 Marks]

Which of the following statements about Dedicated Distributed Storage Service (DSS) is false?

A. DSS disks cannot be expanded while services are running.
B. Backups can be created for disks in the dedicated storage.
C. Three-copy redundancy based on the dedicated distributed storage is available.
D. The capacity of storage pools can be expanded based on service requirements.

Question 8 [8 Marks]

Which of the following statements about FunctionGraph is false?

A. FunctionGraph supports event functions and HTTP functions.
B. You can just write your code and set running conditions without provisioning or managing servers.
C. FunctionGraph supports multiple runtimes, including Node.js, Java, Python, and Go.
D. After a function is created, you cannot edit it inline.

Question 9 [8 Marks]

An enterprise plans to migrate an on-premises Oracle database to Huawei Cloud RDS for MySQL. Which of the following tools is recommended for database schema conversion and data migration?

A. Data Admin Service (DAS)
B. Data Replication Service (DRS)
C. Database Security Service (DBSS)
D. Distributed Database Middleware (DDM)

Question 10 [8 Marks]

Which of the following statements about Object Storage Service (OBS) storage classes is correct?

A. Standard storage class is suitable for storing data that is accessed infrequently.

B. Warm storage class is suitable for real-time data analysis and high-frequency access.

C. Cold (Archive) storage class requires data restoration before access.

D. Deep Archive storage class allows immediate data access without restoration delay.

Question 11 [8 Marks]

Which of the following components in Huawei Cloud Stack provides centralized identity authentication and access permission control?

A. ManageOne

B. IAM

C. Service OM

D. eSight

Question 12 [8 Marks]

An enterprise plans to build a hybrid cloud security system, with some services deployed on Huawei Cloud and others maintained in its local data center. Which of the following statements is false about Huawei Cloud security services?

A. Advanced Anti-DDoS protects only resources on Huawei Cloud. Local data centers cannot be protected.

B. Web Application Firewall (WAF) protects external IDC services through CNAME access.

C. Data Encryption Workshop (DEW) provides unified key management for OBS, EVS, and local storage on the cloud.

D. Host Security Service (HSS) requires an agent installed on the local server to provide intrusion detection.

Question 13 [8 Marks]

A game company develops a global multiplayer online game. Frequently updating gameplay requires dynamic adjustment of data models (for example, adding equipment attributes and season rules). Which of the following Huawei Cloud database services is the most suitable for minimizing the impact of database changes on online players?

A. Database Security Service (DBSS)

B. Distributed Database Middleware (DDM)

C. Document Database Service (DDS)

D. Data Admin Service (DAS)

Question 14 [8 Marks]

Which of the following storage protocols is used by Scalable File Service (SFS) Turbo to provide high-performance shared file storage for compute instances?

A. iSCSI

B. NFS/CIFS

C. Fibre Channel

D. S3 API

Question 15 [8 Marks]

A game company deploys a global server on Huawei Cloud and uses a Layer 4 (TCP/UDP) load balancer to distribute a massive volume of requests. Which of the following describes how a load balancer distributes traffic?

A. SSL/TLS decryption: The load balancer decrypts packets and distributes them based on the IP address of each player.

B. Client certificates are authenticated in an Nginx cluster and then traffic is distributed through HTTP.

C. The load balancer forwards traffic across all nodes in the Nginx cluster and then to backend servers.

D. The load balancer forwards traffic through an LVS cluster and distributes the traffic based on the configured load balancing algorithm.

Question 16 [8 Marks]

A financial institution plans to deploy its core transaction system on Huawei Cloud, with a hybrid cloud architecture. They hope to achieve cross-AZ disaster recovery (DR), RPO = 0 (no data loss), RTO < 15 minutes (quick service recovery), and automatic failover. Which of the following Huawei Cloud backup and DR services should be used?

A. Cloud Server Backup Service (CSBS)

B. Cloud Backup and Recovery (CBR)

C. Volume Backup Service (VBS)

D. Storage Disaster Recovery Service (SDRS)

Question 17 [8 Marks]

A financial enterprise uses Direct Connect to build a hybrid cloud network and plans to configure redundant Direct Connect connections for high availability. Which of the following statements about the connection redundancy solution is false?

A. Automatic failover: If the active Direct Connect connection fails, traffic is automatically routed through the standby connection within one minute.

B. Forced route symmetry: The active and standby Direct Connect connections must use the same BGP routing policy and IP address segment. Otherwise, the switchover will fail.

C. Multiple connection locations: Two Direct Connect connections terminated at different locations in the same region can be used to eliminate single points of failure.

D. Service continuity assurance: Redundant Direct Connect connections can prevent service interruptions caused by unexpected events such as cable cuts.

Question 18 [8 Marks]

A company running an e-commerce application on Huawei Cloud is preparing for a big promotion activity. They want massive user requests to be distributed across multiple backend servers, and they want the number of servers to be automatically scaled in or out as traffic changes. They do not need to manually configure and adjust routing policies. Which of the following combinations of cloud services is the best fit for this scenario?

A. EIP + ELB

B. Elastic Load Balance (ELB) + Auto Scaling

C. NAT Gateway + Auto Scaling

D. Elastic IP (EIP) + Auto Scaling

Question 19 [8 Marks]

Which of the following network services allows Elastic Cloud Servers (ECSs) in a private network to access the Internet without exposing their private IP addresses?

A. EIP
B. NAT Gateway (SNAT)
C. VPC Peering
D. Direct Connect

Question 20 [8 Marks]

A startup plans to deploy a website on Huawei Cloud to showcase its brand. Given the website's low traffic and modest performance needs, which of the following ECS types would be the most cost-effective choice?

A. Storage-intensive
B. General-purpose
C. Memory-intensive
D. Computing-intensive

Question 21 [8 Marks]

A financial enterprise has deployed its core transaction system on Huawei Cloud. Which of the following is the most appropriate way to defend against frequent calls to APIs from malicious crawlers, SQL injection attacks on databases, and sensitive data leaks?

A. CBH
B. HSS
C. WAF
D. AAD

Question 22 [8 Marks]

A video platform plans to expand its cloud storage to support long-term retention of 4K HD video files and enable high-speed access for users worldwide. Which of the following is the most suitable Huawei Cloud service?

A. Elastic Volume Service (EVS)
B. Object Storage Service (OBS)
C. Scalable File Service (SFS)
D. Cloud Backup and Recovery (CBR)

Question 23 [8 Marks]

An e-commerce platform uses Huawei Cloud Elastic Load Balance (ELB) and Auto Scaling to distribute user requests across Elastic Cloud Servers (ECSs) with the same weight in the backend server group. However, monitoring metrics show that the CPU usage on these ECSs differs significantly. What could be the main reason for the load imbalance?

A. HTTP/2 is enabled, so the multiplexing feature changes the request allocation logic.
B. Sticky session is enabled, so the requests from the same user are routed to the same ECS.
C. ECSs with different specifications result in varying vCPU performance.
D. Health check is configured, so traffic is not routed to unhealthy servers, which increases pressure on remaining servers.

Question 24 [8 Marks]

An enterprise uses Huawei Cloud Container Engine (CCE) for containerization, and its service modules only need to communicate internally within the cluster. They do not require external access. Which Service type best supports internal service discovery and load balancing?

A. None

B. LoadBalancer

C. NodePort

D. ClusterIP

Question 25 [8 Marks]

Which of the following is the primary function of Huawei Cloud CodeArts Pipeline?

A. Code hosting and version control

B. Orchestrating and automating end-to-end continuous delivery pipelines

C. Automated UI performance testing

D. Managing container image repositories

Question 26 [8 Marks]

Which of the following database engines is fully compatible with MongoDB on Huawei Cloud?

A. GaussDB(for MySQL)

B. Document Database Service (DDS)

C. Relational Database Service (RDS)

D. Distributed Database Middleware (DDM)

Question 27 [8 Marks]

In Huawei Cloud CCE, which component running on each worker node is responsible for managing the lifecycle of Pods and containers?

A. kube-apiserver

B. kube-controller-manager

C. kubelet

D. etcd

Question 28 [8 Marks]

A company deploys multiple Elastic Cloud Servers (ECSs) on Huawei Cloud, and all these ECSs need to access Internet resources. Which of the following is an action they can take during cloud network planning to reduce their Elastic IP (EIP) costs?

A. Bind an EIP to each ECS.

B. Apply for a NAT proxy server and bind an EIP to it. The proxy server uses source network address translation (SNAT) to allow the ECSs to share the EIP to access the Internet.

C. Apply for a load balancer and bind an EIP to it, so that the load balancer can function as a proxy to allow the ECSs to access the Internet.

D. Apply for a NAT proxy server and bind an EIP to it. The proxy server can then use destination network address translation (DNAT) to allow the ECSs to share the EIP to access the Internet.

Question 29 [8 Marks]

An enterprise plans to deploy a highly available website on Huawei Cloud Container Engine (CCE), including components like the frontend and backend of the site, and the databases it needs to run. Which of the following deployment solutions is the most inappropriate?

A. MySQL deployed via Helm charts: Leveraging community Helm charts enables quick setup of primary/secondary clusters.

B. Log clearance service deployed via CronJobs: Scheduled tasks like log archiving and cleanup can be automated to run daily.

C. Nginx deployed via Deployments: Multiple pods with HPA allow Nginx to efficiently handle frontend traffic.

D. Tomcat deployed via Jobs: The application services run and automatically exit after completion.

Question 30 [8 Marks]

Enterprises deploy Huawei Cloud Content Delivery Network (CDN) so their users can obtain required content from the nearest point of presence (PoP), improving user experience. Which of the following is not a feature of Huawei Cloud CDN?

A. Edge compute power scheduling: CDN analyzes user requests and dynamically allocates GPU compute resources to process large-scale parallel computing tasks.

B. Easy access management: Customers can configure domain names on the visual CDN console and call open APIs to quickly integrate CDN with existing O&M systems.

C. Smart cache: CDN uses AICache to predict content popularity and SSD storage to improve the cache hit ratio.

D. Acceleration optimization: CDN deploys edge PoPs to shorten the content transmission distance, reducing latency and loading pages faster.

Question 31 [8 Marks]

Which of the following is the primary purpose of configuring Security Groups in Huawei Cloud VPC?

A. Route traffic between different VPCs

B. Provide stateful packet filtering and access control for ECS instances

C. Automatically scale backend instances based on traffic

D. Provide public domain name resolution

Question 32 [8 Marks]

An enterprise plans to use Storage Disaster Recovery Service (SDRS) to configure cross-AZ disaster recovery (DR) for critical service systems. Which of the following operation processes is correct?

A. 1. Enable protection. 2. Create protection groups. 3. Create protected instances.

B. 1. Create protection groups. 2. Create protected instances. 3. Enable protection.

C. 1. Enable protection. 2. Create protected instances. 3. Create protection groups.

D. 1. Create protected instances. 2. Create protection groups. 3. Enable protection.

Question 33 [8 Marks]

Which of the following is not a standard cloud computing deployment model?

A. Private cloud

B. Public cloud

C. Hybrid cloud

D. Dedicated cloud

Question 34 [8 Marks]

An enterprise needs to standardize Docker image management during containerization. Which of the following statements is false about the core features of Docker images?

A. Once you have made the necessary modifications to a container, you can use the 'docker commit' command to create a new image.

B. An image consists of multiple read-only layers. The same layer can be shared across images.

C. The latest dependencies are automatically downloaded and installed during image builds.

D. An image contains applications and their complete runtime dependency environments.

Question 35 [8 Marks]

Which of the following network services provides dedicated, high-speed, and secure physical connections between an on-premises data center and Huawei Cloud?

A. VPN

B. Direct Connect

C. VPC Peering

D. Cloud Connect

Question 36 [8 Marks]

Which of the following database services on Huawei Cloud is an enterprise-class distributed relational database based on a cloud-native compute-storage separation architecture?

A. RDS for MySQL

B. GaussDB

C. DDS

D. DDM

Question 37 [8 Marks]

A company uses Huawei Cloud Virtual Private Cloud (VPC) peering connections for cross-network communications. Which of the following statements about the core features of VPC peering connections is false?

A. A VPC peering connection can be created between VPCs in different regions.

B. VPC peering connections between VPCs that have overlapping subnet CIDR blocks may not be usable.

C. Only one VPC peering connection can be created between two VPCs at the same time.

D. A VPC peering connection can connect VPCs from different accounts.

Question 38 [8 Marks]

A financial enterprise plans to use Huawei Cloud Bare Metal Server (BMS) to meet compliance requirements. Which of the following is not one of BMS highlights?

A. Efficient deployment, provisioning within minutes, unified console, open APIs and SDKs, and self-service lifecycle management and O&M

B. Enterprise-grade security, secure and reliable services compliant with regulations

C. Improved deployment density and performance to meet the requirements of mission-critical services such as enterprise databases, big data, containers, high-performance computing (HPC), and AI

D. Manual combination with other cloud services based on the Virtual Private Cloud (VPC) model, changeable VPC for a purchased BMS

Question 39 [8 Marks]

A company plans to use IAM to manage access permissions for their cloud resources. Which of the following statements about IAM core capabilities is true?

A. Resource creation: IAM users can create resources such as VMs and databases without depending on project quotas.

B. Network isolation configuration: IAM policies can be used to define VPC network topologies and subnet division rules for physical isolation between users.

C. Fine-grained permission control: IAM supports RBAC-based API-level permission control for users and user groups.

D. Local identity synchronization: IAM automatically synchronizes all user passwords in the enterprise AD domain controller to the cloud platform.

Question 40 [8 Marks]

Which of the following monitoring services on Huawei Cloud provides real-time metric collection, alarm notification, and resource dashboard views?

A. Log Tank Service (LTS)

B. Cloud Eye

C. Cloud Trace Service (CTS)

D. Application Performance Management (APM)

Question 41 [8 Marks]

A company plans to use cloud computing services to improve IT resource management efficiency. Which of the following statements is true about the core feature of on-demand and self-service usability of cloud computing?

A. Cloud computing service providers only need to prepare the infrastructure resources. Users can complete the remaining work by themselves.

B. Users need to solve all problems by themselves when using cloud computing resources.

C. Users do not need to negotiate with service providers at any stage of using cloud computing resources.

D. Users can generate the cloud computing resource orders they need by themselves.

Question 42 [8 Marks]

An enterprise plans to deploy services on Huawei Cloud Container Engine (CCE). Which of the following is not a typical application scenario of CCE?

A. Traffic management of microservices that are built in containers for capabilities like grayscale release and traffic monitoring

B. E-commerce services that experience traffic surges during activities like promotions and flash sales and require cloud computing resources to be scaled quickly and automatically

C. Live streaming that experiences fluctuating service loads and requires real-time scaling based on CPU or memory usage

D. Core database services in key sectors, like enterprise and finance, that need to be scaled dynamically to ensure efficient operation of services and cross-region high availability

Question 43 [8 Marks]

Which of the following is not a deep learning text classification method?

A. TextCNN

B. AdaBoost

C. TextRNN

D. BERT

Question 44 [8 Marks]

Which of the following machine learning algorithms is an ensemble learning method based on decision trees?

A. K-Means

B. Random Forest

C. Logistic Regression

D. Support Vector Machine (SVM)

Question 45 [8 Marks]

Which of the following tasks is not supported by ModelArts built-in models?

A. Image classification

B. Image detection

C. Gene sequencing

D. Image segmentation

Question 46 [8 Marks]

Which of the following statements about the traditional acoustic model GMM-HMM is false?

A. Third-order HMM models are commonly used in speech recognition tasks.

B. GMM-HMM is an acoustic model. The next step is to use a language model to generate text.

C. A Markov chain, or a Markov process, is a stochastic discrete-event process that satisfies the Markov property. In this process, given the current knowledge or information, any prediction depends solely on the present state. The past is irrelevant.

D. The goal of Markov model is to learn the probabilities of state transitions based on observable data.

Question 47 [8 Marks]

Which of the following Huawei Cloud services is a one-stop AI development platform providing data preprocessing, model training, and deployment?

A. ModelArts

B. HiLens

C. Graph Engine Service (GES)

D. DataArts Studio

Question 48 [8 Marks]

Which of the following is not an application of natural language processing (NLP)?

A. Text-to-Speech (TTS)

B. Machine translation

C. Automatic abstracting

D. Public opinion analysis

Question 49 [8 Marks]

Which of the following is part of the speech synthesis process?

A. Text normalization
B. Text-to-phoneme conversion
C. Waveform generation
D. Pre-emphasis

Question 50 [8 Marks]

Which of the following is not a step in speech recognition preprocessing?

A. Feature normalization
B. Noise reduction
C. Pre-emphasis
D. Edge enhancement

Question 51 [8 Marks]

Which of the following statements about Convolutional Neural Network (CNN) applications in text tasks and image tasks is false?

A. CNN uses convolution kernels to extract local n-gram features in text-related tasks.
B. CNNs work equally well for both images and text.
C. CNNs work better with short texts than long ones.
D. CNNs typically process images as multidimensional pixel matrices.

Question 52 [8 Marks]

Which of the following statements about median filtering in image processing is false?

A. Median filtering reduces salt and pepper noise more effectively than mean filtering.
B. Median filtering improves image clarity and works effectively on pictures with Gaussian noise.
C. It is a filter of the template sorting operation.
D. It replaces the original pixel value with the median output of the pixel values sorted in the neighborhood.

Question 53 [8 Marks]

The softmax function is often used in multiclass classification models in machine learning. Which layer of a deep neural network is the softmax function mainly used for?

A. Input layer
B. Output layer
C. Pooling layer
D. Convolutional layer

Question 54 [8 Marks]

Which of the following statements about overfitting and underfitting is true?

A. Underfitting means that a model performs well during training but poorly during prediction.
B. The underfitting problem can be solved by adding training dataset samples.
C. The degradation of neural networks is not caused by overfitting.
D. Overfitting means that a model does not fully fit its training data.

Question 55 [8 Marks]

Expert systems are a key part of early AI research. To which school of AI thought do they belong?

A. Actionism
B. Bonicsism
C. Connectionism
D. Symbolism

Question 56 [8 Marks]

In what applications do Convolutional Neural Networks (CNNs) excel most?

A. Knowledge graph
B. Computer vision
C. Speech recognition
D. Machine translation

Question 57 [8 Marks]

Which of the following activation functions is non-linear and suffers from the vanishing gradient problem when inputs are very large or very small?

A. ReLU
B. Sigmoid
C. Leaky ReLU
D. ELU

Question 58 [8 Marks]

In deep learning, which of the following processes does ModelArts data management address?

A. Inference
B. Training
C. Data preparation
D. Fine-tuning

Question 59 [8 Marks]

Which of the following is a key advantage of ResNet (Residual Neural Network)?

A. Reduces model size by half
B. Mitigates the vanishing/exploding gradient problem in deep networks using residual connections
C. Eliminates the need for activation functions
D. Enables unsupervised learning on unlabelled images

Question 60 [8 Marks]

Which of the following ModelArts training parameters is used to customize hyperparameters during algorithm tuning?

A. Compute Nodes
B. Resource Pool
C. Algorithm Type
D. Hyperparameter

Question 61 [8 Marks]

Which of the following is the mathematical foundation of the backpropagation algorithm in neural networks?

A. Matrix transposition
B. Chain rule of calculus
C. Fourier transform
D. Monte Carlo simulation

Question 62 [8 Marks]

Which of the following is a feature of the architecture of Huawei Ascend 910 AI chips?

A. MIPS
B. Da Vinci architecture
C. Von Neumann architecture
D. x86

Question 63 [8 Marks]

Which of the following metrics measures the ratio of true positive predictions among all positive predictions made by a model?

A. Recall
B. Precision
C. Accuracy
D. F1-Score

Question 64 [8 Marks]

Which of the following statements about Large Language Model (LLM) training is false?

A. The development cost is high.
B. A single processing unit (PU) usually handles the entire model.
C. There is usually petabyte or terabytes of data involved.
D. There are billions to trillions of parameters.

Question 65 [8 Marks]

Which of the following statements about the function of the update gate in a GRU (Gated Recurrent Unit) is true?

A. It determines the influence of the previous hidden state on the current candidate hidden state.
B. It adjusts the learning rate.
C. It controls the path of gradient backpropagation.
D. It manages long-term dependencies.

Question 66 [8 Marks]

When using a 5 x 5 convolution kernel to perform convolution on a three-channel color image without bias, which of the following is the total number of parameters that need to be learned?

A. 76
B. 75
C. 25
D. 26

Question 67 [8 Marks]

Which AI framework was independently developed by Huawei to support full-scenario AI development across cloud, edge, and device?

A. TensorFlow
B. PyTorch
C. MindSpore
D. Caffe

Question 68 [8 Marks]

Which of the following activation functions ranges from 0 to positive infinity?

A. ELU
B. Tanh
C. ReLU
D. Softmax

Question 69 [8 Marks]

Which of the following compute resources can be selected when you create a training job on ModelArts?

A. CPUs
B. GPUs
C. TPUs
D. NPUs (Ascend)

Question 70 [8 Marks]

Which of the following optimization algorithms uses both momentum and adaptive learning rates for neural network parameter updates?

A. SGD
B. AdaGrad
C. Adam
D. RMSprop

Section 2: Multiple-Answer Questions (Q71 – Q100) [240 Marks]

Question 71 [8 Marks — Select All Correct Options]

In a Kubernetes architecture, which of the following components are deployed on master nodes?

A. API server
B. kubelet
C. Controller manager
D. Scheduler

Question 72 [8 Marks — Select All Correct Options]

An enterprise is deploying a microservice architecture on Huawei Cloud Container Engine (CCE). To ensure functionality, networks must be reachable in every scenario. Which of the following communication modes are supported by Kubernetes?

A. Pods on different nodes communicate directly with each other via pod IP addresses, enabled by Container Network Interface (CNI) plugins, for example, Calico.

B. Kubernetes exposes pods to external networks via Services like NodePort and LoadBalancer, or through ingresses.

C. In the Service-to-pod load balancing scenario, kube-proxy manages iptables or IPVS rules to distribute traffic.

D. Containers within the same pod communicate with each other via localhost and a shared network namespace.

Question 73 [8 Marks — Select All Correct Options]

A financial enterprise uses Huawei Cloud Web Application Firewall (WAF) to protect its online transaction system. Which of the following statements about WAF functions are false?

A. Supported protocols: Security controls, such as WebSocket and SSE, are disabled by default. Customers need to enable and configure these security controls manually.

B. Data masking: WAF can automatically mask fields such as bank card numbers and ID card numbers in attack logs.

C. Precise protection: WAF can protect against CC attacks based on conditions such as geographical locations and HTTP headers.

D. Supported ports: Only standard ports such as 80 and 443 are supported. AAD is required to protect non-standard ports.

Question 74 [8 Marks — Select All Correct Options]

An e-commerce platform plans to use Huawei Cloud Auto Scaling (AS) to handle traffic fluctuations during promotions. Multiple scaling policies need to be configured to intelligently schedule resources. Which of the following policies can be used for automatic scaling?

A. Periodic policy: AS adds 30% more ECSs at 18:00 every Friday to handle weekend traffic peaks.

B. Alarm policy: AS automatically adds two Elastic Cloud Servers (ECSs) when the CPU usage exceeds 80% for 5 minutes.

C. Scheduled policy: AS adds 20 ECSs at 00:00 on November 11.

D. Manual policy: O&M personnel manually adjust the number of ECSs on the console.

Question 75 [8 Marks — Select All Correct Options]

Object Storage Service (OBS) offers a range of resource management tools. Which of the following statements about these tools are false?

A. OBS Browser+ is a graphical Windows client that lets you manage OBS resources from your desktop with ease.

B. obsfs is a GUI tool for accessing and managing OBS resources. It is recommended for batch processing and automated tasks.

C. obsfs is used to mount OBS buckets, but cannot be used to mount OBS parallel file systems.

D. obsutil is a tool based on Filesystem in Userspace (FUSE). It is used to mount OBS parallel file systems to Linux.

Question 76 [8 Marks — Select All Correct Options]

A financial institution plans to migrate its mission-critical databases to Huawei Cloud Relational Database Service (RDS) in compliance with DJCP Multi-level Protection Scheme (MLPS) L3 requirements. Which of the following security capabilities are supported by RDS?

A. Vulnerability scan: The system automatically detects database patch statuses and generates suggestions.

B. Transmission encryption: SSL/TLS 1.2+ is enforced to prevent man-in-the-middle attacks from compromising communications.

C. Network isolation: Instance-level access control is implemented using security groups and subnets.

D. Storage encryption: The Key Management Service (KMS) keys are used to encrypt disk data with AES-256.

Question 77 [8 Marks — Select All Correct Options]

Which of the following are benefits of migrating enterprise applications to Huawei Cloud?

A. Reduced total cost of ownership (TCO) through pay-as-you-go billing

B. Elastic scaling to handle traffic fluctuations

C. High availability with multi-AZ disaster recovery

D. Complete elimination of local application maintenance responsibility

Question 78 [8 Marks — Select All Correct Options]

Which of the following cloud services provide network security protection on Huawei Cloud?

A. Web Application Firewall (WAF)

B. Advanced Anti-DDoS (AAD)

C. Security Group

D. Elastic Load Balance (ELB)

Question 79 [8 Marks — Select All Correct Options]

An AI startup is deploying a machine learning training platform on Huawei Cloud Container Engine (CCE). They require high-performance persistent storage for their workloads. Which of the following Huawei Cloud storage volumes can be directly mounted to a container using a PersistentVolumeClaim (PVC)?

A. Elastic Volume Service (EVS)

B. SFS Turbo

C. Scalable File Service (SFS)

D. Object Storage Service (OBS)

Question 80 [8 Marks — Select All Correct Options]

An enterprise uses Huawei Cloud to deploy services. Their Elastic Cloud Servers (ECSs) are deployed in different Virtual Private Clouds (VPCs) (such as a finance VPC and an R&D; VPC) in the same region. To allow these ECSs to share files across VPCs and meet the low-latency, high-throughput requirements at the same time, which of the following are feasible solutions?

A. Using obsfs to mount Object Storage Service (OBS) parallel file systems to Linux servers

B. Using SFS Turbo and configuring VPC peering connections to enable communications between two or more VPCs in the same region

C. Creating shared Elastic Volume Service (EVS) disks and manually attach them to the ECSs in each VPC

D. Using Scalable File Service (SFS) and configuring multi-VPC access

Question 81 [8 Marks — Select All Correct Options]

Which of the following are core features of Relational Database Service (RDS) on Huawei Cloud?

A. Automated automated backup and point-in-time recovery

B. High availability active/standby deployment

C. Read replica support for read-write splitting

D. Automatic NoSQL schema generation

Question 82 [8 Marks — Select All Correct Options]

O&M; personnel uses Log Tank Service (LTS) to monitor their company system and finds that some logs cannot be viewed on the console. Which of the following are possible causes?

A. Slow browser: There are too many log requests.

B. ICAgent not installed: The log collection client is not deployed on the host and data is not collected.

C. Log group expiration: The log retention duration is set to one day and logs have been automatically deleted.

D. Insufficient IAM permissions: The current account does not have the LTS ReadOnlyAccess policy.

Question 83 [8 Marks — Select All Correct Options]

Which of the following components are part of the Huawei Cloud Stack management platform ManageOne?

A. Operation Center

B. Service Center

C. Service OM

D. eSight

Question 84 [8 Marks — Select All Correct Options]

Which of the following statements are true about Docker core components?

A. OverlayFS uses a tiered storage driver based on the copy-on-write (CoW) strategy to enhance image storage efficiency.

B. containerd is a daemon for container runtime. It manages images and container lifecycle.

C. kubelet is a core component of Kubernetes. It is used to directly create Docker containers.

D. Namespaces isolate resources like processes and networks. They allow each container to have an independent PID, IPC, and network space.

Question 85 [8 Marks — Select All Correct Options]

Which of the following are primary use cases for FunctionGraph on Huawei Cloud?

A. Event-driven data processing (e.g. OBS file upload triggers)

B. Building lightweight RESTful microservice APIs

C. Running long-duration, stateful monolithic enterprise databases

D. Scheduled backend cron tasks

Question 86 [8 Marks — Select All Correct Options]

Which of the following security mechanisms can be used to protect data stored in Object Storage Service (OBS)?

A. Server-side encryption (KMS / SSE-OBS)

B. Bucket ACLs and Bucket Policies

C. Transport encryption via HTTPS/SSL

D. Stateful security group ingress filtering

Question 87 [8 Marks — Select All Correct Options]

An e-commerce platform plans to use Auto Scaling (AS) of Huawei Cloud to handle traffic fluctuations during promotions. Multiple scaling policies need to be configured to intelligently schedule resources. Which of the following are AS policies provided by Huawei Cloud?

A. Alarm policy: AS adds instances automatically when the CPU usage exceeds 80%.
B. Periodic policy: AS adds 50% more instances every Friday to handle weekend traffic.
C. Priority policy: AS removes instances of specific flavors preferentially.
D. Scheduled policy: AS adds two instances at 20:00 every day to handle evening peaks.

Question 88 [8 Marks — Select All Correct Options]

Which of the following are key components of the ModelArts AI development platform?

A. Data Management and Labeling
B. Algorithm Development Environment (JupyterLab)
C. Distributed Model Training
D. Model Deployment & Management (Real-time / Batch)

Question 89 [8 Marks — Select All Correct Options]

Which of the following are optimization methods for foundation model inference?

A. KV caching
B. Pruning
C. Quantization
D. Distillation

Question 90 [8 Marks — Select All Correct Options]

Which of the following are voice processing tasks in AI?

A. Language generation
B. Language understanding
C. Text To Speech (TTS)
D. Speech recognition

Question 91 [8 Marks — Select All Correct Options]

Standard normal distribution is a special form of normal distribution. Which of the following statements about standard normal distribution are true?

A. The variance is 0.
B. The variance is 1.
C. The average value is 0.
D. The average value is 1.

Question 92 [8 Marks — Select All Correct Options]

Which of the following are activation functions of a neural network?

A. ReLU
B. Softplus
C. Sigmoid
D. Softmax

Question 93 [8 Marks — Select All Correct Options]

Dropout is a widely used regularization technique in deep learning. Which of the following are the main functions of dropout?

A. Preventing overfitting
B. Feature extraction
C. Dimension reduction
D. Enhancing model generalization

Question 94 [8 Marks — Select All Correct Options]

Which of the following statements about feature selection methods in text classification are true?

A. Support vector machine (SVM)
B. Information gain
C. Chi-squared test
D. Mutual information

Question 95 [8 Marks — Select All Correct Options]

Which of the following are function modules of CANN (Compute Architecture for Neural Networks)?

A. TBE (Tensor Boost Engine)
B. Underlying compiler
C. FusionEngine
D. CCE operator library

Question 96 [8 Marks — Select All Correct Options]

During neural network training, which of the following tasks are finished before the loss function value is calculated?

A. The model receives input data features and makes predictions by means of forward propagation.
B. The network structure is defined and learnable parameters are initialized.
C. The model reaches convergence and the training ends.
D. Learnable parameters are updated using the gradient descent and backpropagation algorithms to minimize the loss value.

Question 97 [8 Marks — Select All Correct Options]

Which of the following AI engines are supported by ModelArts?

A. MXNet
B. TensorFlow
C. MindSpore
D. PyTorch

Question 98 [8 Marks — Select All Correct Options]

Which of the following statements are true about the advantages of MindSpore?

A. Collaborative optimization with MindSpore and Ascend AI Processors significantly improves operation efficiency and computing performance.

B. MindSpore supports object-oriented and function-oriented programming paradigms, both of which can be used to build network algorithms and training processes.

C. MindSpore natively adapts to each scenario and supports on-demand synergy across devices, edges, and the cloud, as well as flexible deployment.

D. Implementing AI algorithms as code makes the develop time more friendly to developers, significantly reducing the time required for model development and lowering the bar for new developers.

Question 99 [8 Marks — Select All Correct Options]

Which of the following statements about the standard RNN model is false?

A. There is a one-to-one model structure.

B. There is no many-to-many model structure.

C. The time direction element is irrelevant during backpropagation.

D. There is a one-to-many model structure.

Question 100 [8 Marks — Select All Correct Options]

Which of the following are basic image recognition tasks in Computer Vision?

A. Image classification

B. Image segmentation

C. Object detection

D. Object tracking

Section 3: True or False Questions (Q101 – Q150) [200 Marks]

Question 101 [4 Marks]

On Huawei Cloud, an Elastic IP (EIP) can be bound to only one Elastic Cloud Server (ECS) or Load Balancer instance at a time.

[ ] True [ ] False

Question 102 [4 Marks]

In a Virtual Private Cloud (VPC), subnets in the same VPC can communicate with each other by default via the local router.

[ ] True [ ] False

Question 103 [4 Marks]

A security group rule is stateful. If an inbound request is allowed, the corresponding outbound response is automatically allowed regardless of outbound rules.

[ ] True [ ] False

Question 104 [4 Marks]

To ensure the stability of RDS for MySQL read replicas, the specifications (such as vCPUs and memory) of read replicas must be exactly the same as those of the primary DB instance. Or, there may be replication delay or query performance deterioration.

[ ] True [ ] False

Question 105 [4 Marks]

WAF checks TCP/IP requests and provides a variety of functions to help keep web services secure and stable. These functions include asset overview, host management, container management, host fingerprints, container asset management, vulnerability management, baseline checks, container image security, cluster environment security, application protection, web tamper protection, ransomware protection, file integrity management, and virus scanning. Currently, WAF provides cloud, dedicated, shared, and distributed mode to meet diverse security requirements.

[ ] True [ ] False

Question 106 [4 Marks]

Huawei Cloud Object Storage Service (OBS) can use the OBSA-HDFS protocol to directly connect to big data computing frameworks such as Spark and Flink. This allows for HDFS-compatible data storage and access, so big data components can use OBS as a persistent storage layer without requiring architectural changes, thereby fully supporting the data lake architecture.

[ ] True [ ] False

Question 107 [4 Marks]

SFS Turbo provides POSIX-compliant shared file systems for high-concurrency container and HPC scenarios.

[ ] True [ ] False

Question 108 [4 Marks]

Direct Connect provides higher network security and lower latency compared to IPSec VPN over the public Internet.

[ ] True [ ] False

Question 109 [4 Marks]

In Auto Scaling (AS), an AS group can contain instances across multiple Availability Zones (AZs) in the same region.

[ ] True [ ] False

Question 110 [4 Marks]

An online education platform, currently deployed in a Huawei Cloud region in East China, is experiencing high latency for global users. To optimize the experience, a new access point should be established in an international Huawei Cloud region, such as AP-Singapore. This configuration will use the Huawei Cloud backbone network to enable nearby access and intelligent scheduling of requests for reduced cross-border latency.

[ ] True [ ] False

Question 111 [4 Marks]

Cloud Backup and Recovery (CBR) supports crash-consistent and application-consistent backup for ECS instances and EVS volumes.

[ ] True [ ] False

Question 112 [4 Marks]

Cloud Trace Service (CTS) records operations performed on Huawei Cloud resources for security auditing and compliance.

[ ] True [ ] False

Question 113 [4 Marks]

To monitor Java applications, you need to install Application Performance Management (APM) Agents. To monitor Go or Node.js applications, simply configure environment variables on servers. You can use the built-in cloud-init to collect service call data non-intrusively.

[ ] True [ ] False

Question 114 [4 Marks]

Application Performance Management (APM) monitors SQL statement execution. By analyzing metrics such as slow queries and error rates, APM helps locate and optimize the statements that cause service delay or performance bottlenecks.

[ ] True [ ] False

Question 115 [4 Marks]

On the Host Security Service (HSS) console, you can fix the vulnerabilities of all servers across all regions with one click. You do not need to manually select servers or confirm their fix policies.

[ ] True [ ] False

Question 116 [4 Marks]

Data Encryption Workshop (DEW) provides Key Management Service (KMS) for centralized management of encryption keys.

[ ] True [ ] False

Question 117 [4 Marks]

Microservice architecture improves development agility through service splitting and team autonomy. It also significantly simplifies O&M for organizations thanks to multi-service monitoring, distributed transactions, and cross-service link tracing.

[ ] True [ ] False

Question 118 [4 Marks]

ServiceStage supports full-lifecycle management of microservice applications, including build, deployment, monitoring, and governance.

[ ] True [ ] False

Question 119 [4 Marks]

CCE clusters support both Flannel and Calico CNI plugins for container network communication.

[ ] True [ ] False

Question 120 [4 Marks]

Software Repository for Container (SWR) provides secure, reliable container image management and distribution.

[ ] True [ ] False

Question 121 [4 Marks]

GaussDB supports both active/standby and distributed deployment modes to meet varying business concurrency requirements.

[ ] True [ ] False

Question 122 [4 Marks]

In Distributed Database Middleware (DDM), database sharding is completely transparent to application code written for MySQL.

[ ] True [ ] False

Question 123 [4 Marks]

Scalable File Service (SFS) supports cross-VPC access. Users can configure VPC peering connections or use VPC Endpoint (VPCEP) to enable communication between ECSs in different VPCs. They also need to ensure that security group rules allow for cross-VPC traffic. This way, ECSs in different VPCs can share the same file system.

[ ] True [ ] False

Question 124 [4 Marks]

Huawei Cloud CodeArts PerfTest supports real-time test and monitoring and offline report analysis. Users can view key metrics such as concurrency and response time in real time during a test. A detailed PDF report containing data analysis charts (such as TPS and error rate) will be generated after the test.

[ ] True [ ] False

Question 125 [4 Marks]

An enterprise needs to migrate petabytes of data to Huawei Cloud. It is estimated that migration over the public network would take a long time. In this case, Data Express Service (DES) is recommended. DES allows for high-speed offline data migration through offline mailing. This greatly shortens the transfer time.

[ ] True [ ] False

Question 126 [4 Marks]

Key Management Service (KMS) is a secure, reliable, and easy-to-use key hosting service that helps you create, manage, and protect keys. KMS enables server encryption for OBS buckets and stores SSH private key files.

[ ] True [ ] False

Question 127 [4 Marks]

ModelArts supports automated machine learning (AutoML) for zero-code model creation.

[ ] True [ ] False

Question 128 [4 Marks]

In MindSpore, Tensor is the fundamental data structure used to represent multi-dimensional arrays.

[ ] True [ ] False

Question 129 [4 Marks]

Container images are lightweight, executable, independent packages that contain all dependencies and settings required for applications, except for the OS kernel.

[ ] True [ ] False

Question 130 [4 Marks]

Deep learning model training typically requires high-performance compute accelerators such as Ascend NPUs or GPUs.

[ ] True [ ] False

Question 131 [4 Marks]

Deep learning is used for image segmentation, in which classification is used to determine whether a pixel belongs to the foreground or background.

[ ] True [ ] False

Question 132 [4 Marks]

Overfitting occurs when a machine learning model learns noise in the training data, leading to poor generalization on unseen test data.

[ ] True [ ] False

Question 133 [4 Marks]

A color image can be changed into a grayscale image by using a grayscale technique. The color of the grayscale image can deterministically be restored through inverse transformation.

[ ] True [ ] False

Question 134 [4 Marks]

The difference between reinforcement learning and supervised learning is the training objective. The reinforcement signal provided by the environment in reinforcement learning is used to evaluate the action rather than telling the learning system how to perform correct actions.

[ ] True [ ] False

Question 135 [4 Marks]

Backpropagation minimizes the loss function by calculating its gradients with respect to model parameters and iteratively updating those parameters based on the computed gradients.

[ ] True [ ] False

Question 136 [4 Marks]

The deeper a deep neural network is, the better its performance and higher its accuracy is.

[ ] True [ ] False

Question 137 [4 Marks]

Softmax activation function converts raw logits into a probability distribution summing to 1.0.

[ ] True [ ] False

Question 138 [4 Marks]

The lower the value on the diagonal of the confusion matrix is, the better the classification of the classifier is.

[ ] True [ ] False

Question 139 [4 Marks]

The gradient descent algorithm updates parameters to find the minimum point of the loss function. Its convergence point is determined solely by the cost function.

[ ] True [ ] False

Question 140 [4 Marks]

AI matches or exceeds human performance in areas like image recognition and gaming strategies but falls short in tasks like speech recognition and reading comprehension compared to typical human abilities.

[ ] True [ ] False

Question 141 [4 Marks]

Convolutional layers in CNNs use weight sharing to reduce the number of learnable parameters.

[ ] True [ ] False

Question 142 [4 Marks]

When labeling a dataset on ModelArts, you can view the labeling progress, number of labeled images, and total number of images.

[ ] True [ ] False

Question 143 [4 Marks]

Recurrent Neural Networks (RNNs) are specifically designed to handle sequential time-series data or natural language text.

[ ] True [ ] False

Question 144 [4 Marks]

Transformer architectures rely on self-attention mechanisms rather than recurrent connections to capture long-range dependencies.

[ ] True [ ] False

Question 145 [4 Marks]

The probability of an n-gram is estimated based on its frequency of occurrence within the text.

[ ] True [ ] False

Question 146 [4 Marks]

Word2Vec embeds words into a continuous vector space where semantically similar words are close to each other.

[ ] True [ ] False

Question 147 [4 Marks]

RNN is a type of neural network that captures dynamic information in sequential data through periodic connections of hidden layer nodes.

[ ] True [ ] False

Question 148 [4 Marks]

The perceptron assigns a unique weight to each input signal. It multiplies inputs by their corresponding weights, sums these products, and compares the total to a threshold value to decide if the neuron activates.

[ ] True [ ] False

Question 149 [4 Marks]

Ascend 910 processor is designed for high-performance AI model training.

[ ] True [ ] False

Question 150 [4 Marks]

Without an activation function, the neural network remains linear. Adding an activation function transforms it into a non-linear model.

[ ] True [ ] False

PART 2: MASTER ANSWER KEY & SOLUTIONS GUIDE

Complete Answer Matrix & Official Huawei Study Notes for Q1 – Q150

Master Answer Key Quick Reference Matrix

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
1: C	2: C	3: A	4: D	5: D	6: D	7: A	8: D	9: B	10: C
11: B	12: A	13: C	14: B	15: D	16: D	17: B	18: B	19: B	20: B
21: C	22: B	23: B	24: D	25: B	26: B	27: C	28: B	29: D	30: A
31: B	32: B	33: D	34: C	35: B	36: B	37: A	38: D	39: C	40: B
41: D	42: D	43: B	44: B	45: C	46: A	47: A	48: A	49: C	50: D
51: B	52: B	53: B	54: C	55: D	56: B	57: B	58: C	59: B	60: D
61: B	62: B	63: B	64: B	65: D	66: B	67: C	68: C	69: D	70: C
71: A, C, D	72: A, B, C, D	73: A, D	74: A, B, C	75: B, C, D	76: B, C, D	77: A, B, C	78: A, B, C	79: A, B, C, D	80: A, B, D
81: A, B, C	82: B, C, D	83: A, B	84: A, B, D	85: A, B, D	86: A, B, C	87: A, B, D	88: A, B, C, D	89: A, B, C, D	90: C, D
91: B, C	92: A, B, C, D	93: A, D	94: B, C, D	95: A, B, C, D	96: A, B	97: A, B, C, D	98: A, B, C, D	99: A, B, C	100: A, B, C, D
101: True	102: True	103: True	104: False	105: True	106: True	107: True	108: True	109: True	110: True
111: True	112: True	113: False	114: True	115: False	116: True	117: False	118: True	119: True	120: True
121: True	122: True	123: True	124: True	125: True	126: False	127: True	128: True	129: True	130: True
131: True	132: True	133: False	134: True	135: True	136: False	137: True	138: False	139: False	140: False
141: True	142: True	143: True	144: True	145: True	146: True	147: True	148: True	149: True	150: True

Detailed Solutions & Official Study Notes

Section 1: Single-Answer Questions Solutions (Q1 – Q70)

Q1. Correct Answer: C

Official Note: Destination IP address load balancing is not supported in Huawei Cloud ELB; supported algorithms include weighted round robin, source IP hash, and weighted least connections.

Q2. Correct Answer: C

Official Note: VPC peering connections can connect VPCs belonging to different accounts or the same account, provided they are in the same region and have non-overlapping CIDR blocks.

Q3. Correct Answer: A

Official Note: Buying a new VPN connection creates unnecessary operational overhead and cost when VPC peering or Direct Connect sharing via local subnets/routes can reuse the existing physical line.

Q4. Correct Answer: D

Official Note: Cloud Connect supports connecting VPCs across different regions AND across different accounts via cross-account authorization.

Q5. Correct Answer: D

Official Note: A listener in Huawei Cloud ELB can be associated with multiple backend server groups using Layer 7 forwarding rules based on URL paths or domain names.

Q6. Correct Answer: D

Official Note: Virtual Machines (VMs) provide hypervisor-level physical and kernel isolation, whereas containers share the host OS kernel.

Q7. Correct Answer: A

Official Note: DSS disks support online capacity expansion without requiring service shutdown.

Q8. Correct Answer: D

Official Note: FunctionGraph allows inline code editing directly in the web console for supported runtimes.

Q9. Correct Answer: B

Official Note: Data Replication Service (DRS) is Huawei Cloud's primary tool for real-time data migration, schema conversion, and synchronization.

Q10. Correct Answer: C

Official Note: Cold (Archive) storage class in OBS requires restoring data to a readable state before it can be downloaded or accessed.

Q11. Correct Answer: B

Official Note: Identity and Access Management (IAM) provides centralized identity authentication, user permissions, and resource access control across regions.

Q12. Correct Answer: A

Official Note: Advanced Anti-DDoS protects both Huawei Cloud resources and external/on-premises data centers via BGP routing and IP redirection.

Q13. Correct Answer: C

Official Note: Document Database Service (DDS) is a MongoDB-compatible NoSQL database ideal for flexible, schemaless gameplay data models.

Q14. Correct Answer: B

Official Note: SFS Turbo provides shared file systems accessible via NFSv3/NFSv4 and CIFS protocols for high-concurrency access.

Q15. Correct Answer: D

Official Note: Layer 4 ELB forwards TCP/UDP traffic directly at the transport layer using an LVS cluster for high throughput.

Q16. Correct Answer: D

Official Note: Storage Disaster Recovery Service (SDRS) provides cross-AZ block storage replication achieving RPO = 0 and automatic failover.

Q17. Correct Answer: B

Official Note: Direct Connect redundant connections do not force identical BGP routing policies; active-standby or active-active policies can be customized.

Q18. Correct Answer: B

Official Note: Combining Elastic Load Balance (ELB) with Auto Scaling (AS) allows automatic adjustment of ECS instance count and dynamic traffic distribution.

Q19. Correct Answer: B

Official Note: NAT Gateway (SNAT) enables ECSs in private subnets to access the Internet securely without requiring individual public EIPs.

Q20. Correct Answer: B

Official Note: General-purpose ECS instances balance compute, memory, and networking resources cost-effectively for low-traffic websites.

Q21. Correct Answer: C

Official Note: Web Application Firewall (WAF) inspects Layer 7 HTTP/HTTPS traffic to defend against web crawlers, SQL injection, XSS, and data leaks.

Q22. Correct Answer: B

Official Note: Object Storage Service (OBS) provides scalable, low-cost object storage for 4K video files combined with CDN for high-speed delivery.

Q23. Correct Answer: B

Official Note: Sticky session routes all requests from a given client session to the same backend server, causing CPU load imbalances.

Q24. Correct Answer: D

Official Note: ClusterIP is the default Kubernetes Service type for internal service discovery and load balancing within the cluster.

Q25. Correct Answer: B

Official Note: CodeArts Pipeline automates build, test, and deployment stages across the software delivery lifecycle.

Q26. Correct Answer: B

Official Note: Document Database Service (DDS) is fully compatible with MongoDB protocol standards.

Q27. Correct Answer: C

Official Note: kubelet runs on every worker node in a CCE/Kubernetes cluster to manage Pod container lifecycles via the container runtime.

Q28. Correct Answer: B

Official Note: A NAT Gateway with SNAT allows multiple ECSs to share a single public EIP for outgoing Internet traffic, significantly reducing cost.

Q29. Correct Answer: D

Official Note: Tomcat application servers run continuous web services and should be deployed via Deployments, not short-lived Kubernetes Jobs that exit upon completion.

Q30. Correct Answer: A

Official Note: CDN accelerates static and dynamic content delivery via edge PoPs but does not schedule GPU parallel compute tasks.

Q31. Correct Answer: B

Official Note: Security Groups act as virtual firewalls providing stateful packet inspection and access control rules for ECS instances.

Q32. Correct Answer: B

Official Note: The correct SDRS setup process is: 1. Create protection groups, 2. Create protected instances, 3. Enable protection.

Q33. Correct Answer: D

Official Note: Dedicated cloud is a deployment option within public cloud platforms, not one of the three primary cloud models (Private, Public, Hybrid).

Q34. Correct Answer: C

Official Note: Docker images are immutable layered binaries; dependencies are baked in at build time and not downloaded dynamically during container run.

Q35. Correct Answer: B

Official Note: Direct Connect provides dedicated physical fiber lines between on-premises IDCs and Huawei Cloud VPCs.

Q36. Correct Answer: B

Official Note: GaussDB is Huawei Cloud's cloud-native distributed enterprise database with compute-storage separation.

Q37. Correct Answer: A

Official Note: VPC Peering connections are restricted to VPCs within the same region; cross-region interconnections require Cloud Connect.

Q38. Correct Answer: D

Official Note: BMS provides dedicated physical servers without virtualization overhead, fully integrated with VPC networks without manual wiring.

Q39. Correct Answer: C

Official Note: IAM supports fine-grained, Role-Based Access Control (RBAC) and policy-based permission control at the API level.

Q40. Correct Answer: B

Official Note: Cloud Eye is the core O&M; monitoring service for metrics, alarms, and dashboards across Huawei Cloud resources.

Q41. Correct Answer: D

Official Note: On-demand self-service allows users to provision and configure cloud resources autonomously via online consoles or APIs.

Q42. Correct Answer: D

Official Note: Core relational databases are typically deployed on RDS/GaussDB or BMS rather than containerized in CCE due to stateful disk persistence and MLPS L3 requirements.

Q43. Correct Answer: B

Official Note: AdaBoost is a traditional ensemble boosting algorithm, whereas TextCNN, TextRNN, and BERT are deep learning architectures.

Q44. Correct Answer: B

Official Note: Random Forest constructs an ensemble of decision trees to improve classification/regression accuracy and prevent overfitting.

Q45. Correct Answer: C

Official Note: ModelArts built-in models focus on vision, speech, and NLP tasks rather than specialized biological gene sequencing.

Q46. Correct Answer: A

Official Note: Traditional GMM-HMM acoustic models use first-order (1st-order) Markov chains where state transition depends only on the immediate preceding state.

Q47. Correct Answer: A

Official Note: ModelArts is Huawei Cloud's full-stack, one-stop AI development platform spanning data processing, training, and deployment.

Q48. Correct Answer: A

Official Note: TTS (Text-to-Speech) is a speech processing task, whereas machine translation, abstracting, and sentiment analysis are NLP tasks.

Q49. Correct Answer: C

Official Note: Speech synthesis converts text to phonemes and uses neural acoustic/vocoder models for waveform generation.

Q50. Correct Answer: D

Official Note: Edge enhancement is an image processing filtering technique, not a speech signal preprocessing step.

Q51. Correct Answer: B

Official Note: CNNs excel at grid-like 2D spatial image structures; for sequential text, RNNs/Transformers generally outperform standard 1D CNNs.

Q52. Correct Answer: B

Official Note: Median filtering excels at removing impulse (salt-and-pepper) noise, while Gaussian smoothing filters are better for Gaussian noise.

Q53. Correct Answer: B

Official Note: Softmax is applied at the final output layer of multiclass classification networks to normalize raw logits into class probabilities.

Q54. Correct Answer: C

Official Note: Neural network degradation (accuracy saturation in deep networks) is caused by optimization difficulties/vanishing gradients, not overfitting; ResNet solves it with shortcut connections.

Q55. Correct Answer: D

Official Note: Expert systems use explicit rule bases and inference engines, belonging to the Symbolism (symbolic AI) school.

Q56. Correct Answer: B

Official Note: Convolutional Neural Networks (CNNs) excel at spatial pattern extraction in Computer Vision applications.

Q57. Correct Answer: B

Official Note: Sigmoid saturates at 0 and 1, causing zero gradients and the vanishing gradient problem in deep feedforward architectures.

Q58. Correct Answer: C

Official Note: ModelArts data management handles data ingestion, labeling, cleaning, and dataset version preparation.

Q59. Correct Answer: B

Official Note: ResNet introduces residual identity skip connections that allow gradients to flow directly back, solving vanishing gradients in deep networks.

Q60. Correct Answer: D

Official Note: ModelArts allows configuring hyperparameter ranges and tuning policies during training job setup.

Q61. Correct Answer: B

Official Note: Backpropagation computes partial derivatives of the loss function with respect to weights using the calculus chain rule.

Q62. Correct Answer: B

Official Note: Huawei Ascend 910 AI processors are built on Huawei's proprietary Da Vinci architecture.

Q63. Correct Answer: B

Official Note: Precision = True Positives / (True Positives + False Positives), measuring exactness among positive predictions.

Q64. Correct Answer: B

Official Note: Large Language Model (LLM) training requires distributed multi-node parallel computing (Data, Model, and Pipeline Parallelism) across thousands of NPUs/GPUs.

Q65. Correct Answer: D

Official Note: The update gate in GRU determines how much of the previous hidden state is retained vs updated, managing long-term dependencies.

Q66. Correct Answer: B

Official Note: For 3 input channels and a 5x5 kernel without bias: 5 * 5 * 3 = 75 parameters.

Q67. Correct Answer: C

Official Note: MindSpore is Huawei's open-source, all-scenario AI computing framework.

Q68. Correct Answer: C

Official Note: ReLU output ranges from 0 to +infinity (max(0, x)).

Q69. Correct Answer: D

Official Note: ModelArts leverages Huawei Ascend NPUs (alongside CPUs and GPUs) for accelerated model training.

Q70. Correct Answer: C

Official Note: Adam (Adaptive Moment Estimation) combines the benefits of Momentum and RMSprop (adaptive learning rates per parameter).

Section 2: Multiple-Answer Questions Solutions (Q71 – Q100)

Q71. Correct Answers: A, C, D

Official Note: Master nodes in Kubernetes run control plane components: kube-apiserver, kube-controller-manager, and kube-scheduler. kubelet runs on worker nodes.

Q72. Correct Answers: A, B, C, D

Official Note: Kubernetes supports flat Pod-to-Pod CNI IP networking, external Service exposure (NodePort/LoadBalancer/Ingress), iptables/IPVS load balancing via kube-proxy, and localhost container sharing within Pods.

Q73. Correct Answers: A, D

Official Note: WAF supports WebSocket/SSE by default without extra manual toggles, and supports non-standard HTTP/HTTPS ports directly without requiring AAD.

Q74. Correct Answers: A, B, C

Official Note: Auto Scaling supports periodic, alarm-triggered, and scheduled scaling policies. Manual adjustment is a manual administrative action, not an automated policy.

Q75. Correct Answers: B, C, D

Official Note: obsfs is a CLI-based FUSE tool (not a GUI tool) used to mount OBS parallel file systems to Linux instances. obsutil is a CLI utility for batch object operations.

Q76. Correct Answers: B, C, D

Official Note: RDS supports SSL/TLS transmission encryption, Security Group/subnet network isolation, and KMS AES-256 disk storage encryption. Automated patch vulnerability scanning is managed at the platform level.

Q77. Correct Answers: A, B, C

Official Note: Cloud migration reduces TCO, enables auto-scaling, and provides multi-AZ HA. Maintenance of application logic and guest OS/database configurations remains with the customer.

Q78. Correct Answers: A, B, C

Official Note: WAF (Layer 7), Advanced Anti-DDoS (Layer 3/4), and Security Groups (instance firewalls) provide network security protection.

Q79. Correct Answers: A, B, C, D

Official Note: CCE containers can mount EVS block volumes, SFS Turbo file systems, SFS shared files, and OBS buckets via Kubernetes PVCs.

Q80. Correct Answers: A, B, D

Official Note: SFS Turbo with VPC Peering, obsfs with OBS, and multi-VPC SFS allow cross-VPC file sharing. Standard EVS disks cannot be attached across different VPCs without shared EVS setup constraints.

Q81. Correct Answers: A, B, C

Official Note: RDS provides automated point-in-time backups, active/standby HA failover, and read replicas for read/write splitting.

Q82. Correct Answers: B, C, D

Official Note: Missing ICAgent, log group retention expiration, and missing IAM read permissions prevent viewing logs in LTS console.

Q83. Correct Answers: A, B

Official Note: ManageOne consists of two primary operational portals: Operation Center (for business management) and Service Center (for self-service provisioning).

Q84. Correct Answers: A, B, D

Official Note: OverlayFS uses CoW tiered storage, containerd manages runtime lifecycles, and Namespaces isolate process/network environments. kubelet communicates with container runtimes, not directly creating containers.

Q85. Correct Answers: A, B, D

Official Note: FunctionGraph excels at event-driven execution, lightweight microservice APIs, and scheduled cron jobs. Long-running monolithic databases are unsuitable for serverless functions.

Q86. Correct Answers: A, B, C

Official Note: OBS security features include KMS server-side encryption, Bucket ACLs/Policies, and HTTPS transport encryption. Security Groups apply to ECS/instances, not OBS directly.

Q87. Correct Answers: A, B, D

Official Note: Auto Scaling provides alarm-based, periodic, and scheduled scaling policies.

Q88. Correct Answers: A, B, C, D

Official Note: ModelArts provides an end-to-end suite: Data Management, JupyterLab Dev, Distributed Training, and Multi-mode Deployment.

Q89. Correct Answers: A, B, C, D

Official Note: KV caching, model pruning, weight quantization, and knowledge distillation are core foundation model inference optimization techniques.

Q90. Correct Answers: C, D

Official Note: Speech Recognition (ASR) and Text-to-Speech (TTS) are speech/voice processing tasks. Language generation and understanding belong to NLP.

Q91. Correct Answers: B, C

Official Note: A standard normal distribution has a mean (average) of 0 and a variance of 1: N(0, 1).

Q92. Correct Answers: A, B, C, D

Official Note: ReLU, Softplus, Sigmoid, and Softmax are all standard neural network activation functions.

Q93. Correct Answers: A, D

Official Note: Dropout randomly deactivates neurons during training to prevent co-adaptation, mitigating overfitting and improving model generalization.

Q94. Correct Answers: B, C, D

Official Note: Information Gain, Chi-squared test, and Mutual Information are statistical feature selection methods in text classification. SVM is a classifier model.

Q95. Correct Answers: A, B, C, D

Official Note: CANN architecture includes Tensor Boost Engine (TBE), underlying compiler, FusionEngine, and CCE operator acceleration library.

Q96. Correct Answers: A, B

Official Note: Forward propagation prediction generation and parameter initialization precede loss calculation. Loss value calculation is required BEFORE gradient descent parameter updates occur.

Q97. Correct Answers: A, B, C, D

Official Note: ModelArts natively supports MindSpore, TensorFlow, PyTorch, and MXNet training engines.

Q98. Correct Answers: A, B, C, D

Official Note: MindSpore offers hardware-software synergy with Ascend chips, supports functional/OO paradigms, device-edge-cloud synergy, and developer-friendly coding.

Q99. Correct Answers: A, B, C

Official Note: Standard RNNs naturally support sequence-to-sequence (many-to-many), one-to-many, and many-to-one mapping structures, and use Backpropagation Through Time (BPTT) where time steps are fundamental.

Q100. Correct Answers: A, B, C, D

Official Note: Image classification, semantic segmentation, object detection, and object tracking are core computer vision tasks.

Section 3: True or False Questions Solutions (Q101 – Q150)

Q101. Correct Answer: True

Official Note: An EIP can only be bound to a single ECS instance or Load Balancer network interface at any given time.

Q102. Correct Answer: True

Official Note: Subnets inside the same Virtual Private Cloud (VPC) can communicate with each other by default through the underlying VPC virtual router.

Q103. Correct Answer: True

Official Note: Security groups are stateful; return traffic for an allowed incoming request is automatically permitted regardless of outbound rules.

Q104. Correct Answer: False

Official Note: RDS for MySQL read replicas can use different specifications (e.g. higher or lower CPU/memory) than the primary instance based on read workload demands.

Q105. Correct Answer: True

Official Note: Huawei Cloud WAF offers comprehensive security features across cloud, dedicated, and distributed modes.

Q106. Correct Answer: True

Official Note: OBSA-HDFS allows big data engines (Spark, Flink, Hadoop) to access OBS directly as a decoupled data lake storage layer.

Q107. Correct Answer: True

Official Note: SFS Turbo delivers POSIX-compliant, low-latency shared file storage optimized for container clusters and high-performance computing.

Q108. Correct Answer: True

Official Note: Direct Connect bypasses the public Internet via private leased lines, ensuring lower latency and superior security compared to IPSec VPN.

Q109. Correct Answer: True

Official Note: Auto Scaling groups can span multiple AZs within a single region to ensure cross-AZ high availability.

Q110. Correct Answer: True

Official Note: Deploying international edge PoPs and leveraging Huawei Cloud's global backbone network significantly reduces cross-border latency.

Q111. Correct Answer: True

Official Note: CBR provides crash-consistent backups and application-consistent backups using volume shadow copy services (VSS).

Q112. Correct Answer: True

Official Note: CTS captures and logs API calls and user operations across Huawei Cloud for compliance and security auditing.

Q113. Correct Answer: False

Official Note: APM agents or bytecode injection are required for deep Java application call tracing; simple environment variables via cloud-init are insufficient.

Q114. Correct Answer: True

Official Note: APM tracks slow SQL query execution metrics to help developers pinpoint database performance bottlenecks.

Q115. Correct Answer: False

Official Note: HSS vulnerability fixing requires administrator confirmation and selection of target servers to prevent unexpected service restarts.

Q116. Correct Answer: True

Official Note: Data Encryption Workshop (DEW) includes KMS for key management and hardware security module (HSM) protection.

Q117. Correct Answer: False

Official Note: Microservices increase development agility but increase operational and monitoring complexity due to distributed tracing and network coordination.

Q118. Correct Answer: True

Official Note: ServiceStage provides full-lifecycle microservice management from code compilation to deployment, scaling, and governance.

Q119. Correct Answer: True

Official Note: CCE supports both Flannel (overlay) and Calico (network policy & BGP) network plugins.

Q120. Correct Answer: True

Official Note: SWR is Huawei Cloud's enterprise container image registry for managing Docker/OCI images.

Q121. Correct Answer: True

Official Note: GaussDB supports active/standby HA mode and distributed sharded deployment for massive concurrency.

Q122. Correct Answer: True

Official Note: DDM handles database sharding and SQL routing transparently for MySQL-compatible applications.

Q123. Correct Answer: True

Official Note: SFS supports cross-VPC file system mounting via VPC peering connections or VPC Endpoint (VPCEP).

Q124. Correct Answer: True

Official Note: CodeArts PerfTest provides real-time concurrency/TPS metric tracking and auto-generates comprehensive performance test PDF reports.

Q125. Correct Answer: True

Official Note: Data Express Service (DES) provides offline physical storage device mailing for petabyte-scale data migration.

Q126. Correct Answer: False

Official Note: KMS manages cryptographic keys for server-side encryption; SSH private key management is handled by Key Pair Service (KPS).

Q127. Correct Answer: True

Official Note: ModelArts ExModel / AutoML supports automated feature engineering, model search, and hyperparameter tuning without coding.

Q128. Correct Answer: True

Official Note: Tensor is the foundational data structure in MindSpore representing N-dimensional matrices.

Q129. Correct Answer: True

Official Note: Container images include application binaries, system libraries, and configs, sharing the underlying host Linux kernel.

Q130. Correct Answer: True

Official Note: Deep neural network training relies heavily on hardware accelerators like Ascend NPUs or GPUs for matrix multiplication.

Q131. Correct Answer: True

Official Note: Image segmentation performs pixel-level classification to distinguish target objects from background pixels.

Q132. Correct Answer: True

Official Note: Overfitting occurs when a model memorizes training noise, failing to generalize to unseen test samples.

Q133. Correct Answer: False

Official Note: Grayscaling discards chrominance (color) channels; original colors cannot be deterministically restored without generative estimation.

Q134. Correct Answer: True

Official Note: Reinforcement learning uses scalar reward signals from the environment to evaluate actions rather than ground-truth target outputs.

Q135. Correct Answer: True

Official Note: Backpropagation computes gradients via the chain rule and updates weights via gradient descent to minimize loss.

Q136. Correct Answer: False

Official Note: Increasing network depth beyond a certain point causes vanishing/exploding gradients or degradation unless shortcut connections (ResNet) are used.

Q137. Correct Answer: True

Official Note: Softmax normalizes raw output logits into a valid probability distribution where all elements sum to 1.0.

Q138. Correct Answer: False

Official Note: In a confusion matrix, diagonal entries represent correct predictions (True Positives / True Negatives); higher diagonal values indicate better classification.

Q139. Correct Answer: False

Official Note: Gradient descent convergence is heavily influenced by learning rate, initialization, batch size, and optimizer choice, not solely the cost function.

Q140. Correct Answer: False

Official Note: Modern AI equals or surpasses human abilities in speech recognition and reading comprehension, but challenges remain in general common-sense reasoning.

Q141. Correct Answer: True

Official Note: CNNs share convolution kernel weights across spatial locations, drastically reducing parameter count compared to fully connected layers.

Q142. Correct Answer: True

Official Note: ModelArts dataset management displays progress bars, labeled sample counts, and dataset statistics.

Q143. Correct Answer: True

Official Note: RNNs process data sequentially, making them ideal for time-series and natural language sequence modeling.

Q144. Correct Answer: True

Official Note: Transformers replace recurrence with multi-head self-attention mechanisms to process all sequence positions in parallel.

Q145. Correct Answer: True

Official Note: N-gram language models estimate sequence probability based on relative frequency counts in corpus training text.

Q146. Correct Answer: True

Official Note: Word2Vec generates dense continuous vector embeddings where geometric distance reflects semantic similarity.

Q147. Correct Answer: True

Official Note: RNNs maintain hidden state memory through recurrent connections across sequential time steps.

Q148. Correct Answer: True

Official Note: The classic Rosenblatt perceptron computes a weighted sum of inputs and applies a step threshold function.

Q149. Correct Answer: True

Official Note: Huawei Ascend 910 is a high-performance AI processor engineered specifically for training complex deep learning models.

Q150. Correct Answer: True

Official Note: Without non-linear activation functions, stacked neural network layers collapse into a single linear transformation.